CHAPTER 1

Introduction to Analog Design

Mostly about motivation for analog, some interesting notes:

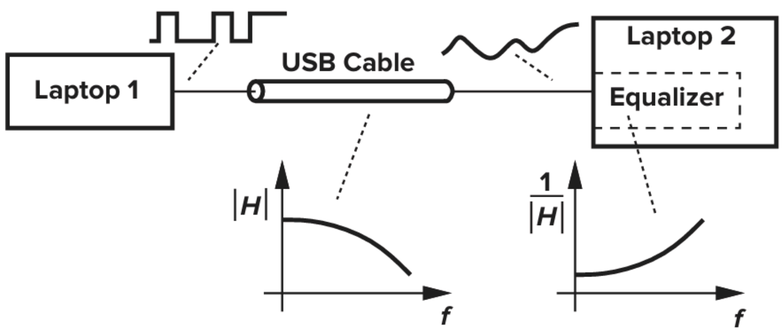
- At low BW it's more efficient to digitalise and process in the digital domain, at higher BW it's more efficient to process in the analog domain. Despite analog components burning more power in general
  - Channel equalisation is an example of the processing
    - 

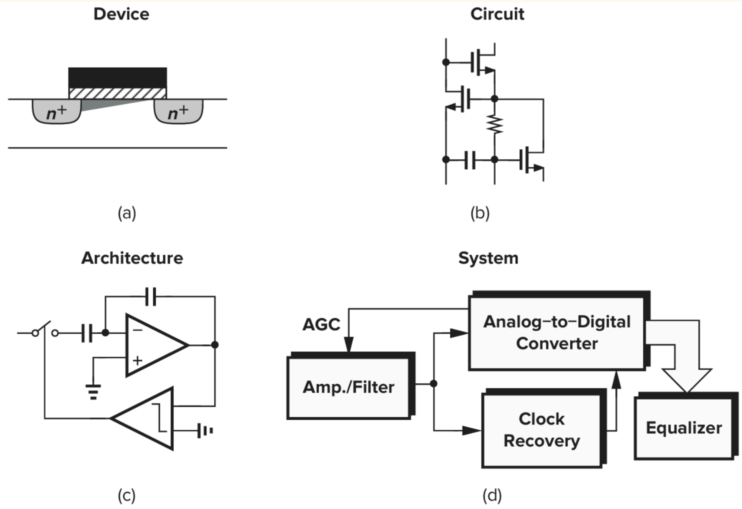
Figure 1.3 Equalization to compensate for high-frequency attenuation in a USB cable.
  - amplifiers, ADCs, and the RF transmitter burns the most power
- MOSFET is nosier than BJT, but can run at lower VDD, 1V vs 2V
- Different levels of abstractions that all needs to be considered
  - 

Figure 1.5 Abstraction levels in circuit design: (a) device level, (b) circuit level, (c) architecture level, (d) system level.